AICW GENAI COMPANION

Generative AI & Prompt Engineering

A plain-English guide to how machines create text, images, code, and conversation

Chapter 7 · Read alongside the lecture slides · Every slide example worked out in full

How to use this companion. Five colours guide you through this reader. A blue box (☞) gives the plain-word idea before any jargon. An amber box (★) gives an everyday analogy. A green box (✎) works a real example step by step. A red box (✗) flags a common mistake. A purple box (✔) gives the one line to remember. Read them in the order they appear and you meet every hard idea with a soft landing first.

Contents

1 The big idea: what is generative AI? 2

2 A quick timeline: how generative AI got here 4

3 Under the hood: the stages of a generative AI system 5

4 Generative AI at work: applications 7

5 Prompt engineering: talking to AI the right way 9

6 Three techniques that make a prompt better 10

7 Five ways to shape a prompt 11

8 Guardrails: ethics and pitfalls of generative AI 13

Glossary & key terms 14

1 The big idea: what is generative AI?

Type one sentence and a computer writes you a poem, draws you a picture, or writes working code. That is generative AI at work. This section shows what makes it different from the AI you may already know.

☞ The intuition

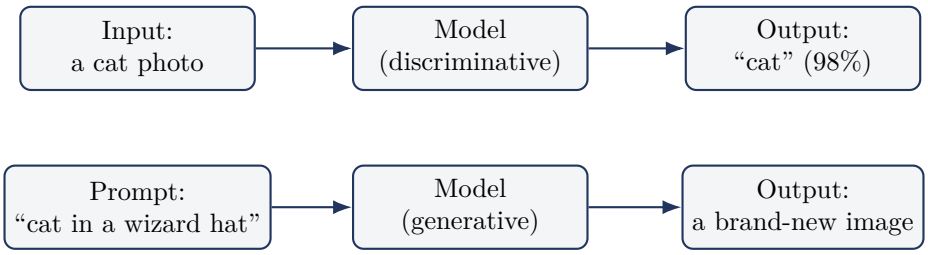
Most AI you meet day to day is a **judge**: show it a photo and it says ”cat” or ”dog.” Generative AI is a **maker**: give it a prompt (an instruction) and it builds something brand new – text, an image, audio, or code – that did not exist before.

★ Everyday picture

Picture a judge at a baking contest next to a baker. The judge (traditional AI) tastes a cake and gives it a score. The baker (generative AI) takes a recipe card, your prompt, and bakes a brand-new cake from scratch. Break point: a baker still needs real ingredients. A generative model with no training data has nothing to bake with.

Generative AI is a type of artificial intelligence that creates new content: text, images, or audio. It learns from huge piles of existing examples, called *training data*, using *deep learning* – a method built from layered neural networks. Once trained, it can produce brand-new content that fits the style of what it learned from.

This sits opposite a **discriminative model**, which sorts input into an already-known category, such as ”spam” or ”not spam.” A discriminative model never makes anything new; it only chooses from a fixed list of labels.



How to read it: Top row: a discriminative model only ever picks a label from a fixed list. Bottom row: a generative model builds new content it has never seen before, guided by the prompt.

✎ Worked example: traditional AI vs. generative AI, side by side

Suppose we feed a photo of a cat into both kinds of system.

Step 1. Traditional (discriminative) AI reads the photo. It has one job: classify.

Step 2. It compares the photo to categories it learned. These are fixed labels: ”cat,” ”dog,” ”bird.”

Step 3. It outputs one label plus a confidence score. For example: ”cat, 98% confident.”

Step 4. Generative AI, same photo, a different job. We ask: ”Draw this cat wearing a wizard hat, in watercolor style.”

Step 5. It does not choose from a fixed list. It builds a brand-new image, piece by piece, that never existed before.

Step 6. Sense-check. The first system could never draw a picture – it only sorts.

The second could never grade a school quiz – it only creates. They solve opposite problems.

Where you'll meet this in real tools. ChatGPT, DALL · E, and GitHub Copilot are generative – they make new text, images, or code. A spam filter or a medical-scan classifier is discriminative – it only sorts what already exists into known buckets.

✗ Watch out

People often assume "AI" always means "smart judge." Many everyday AI tools, like spam filters and fraud detectors, are discriminative, not generative. Do not assume every AI system can create new content – check which kind you are actually using.

✔ Key takeaway

Generative AI does not just sort what exists – it makes something new, in the style of what it learned.

2 A quick timeline: how generative AI got here

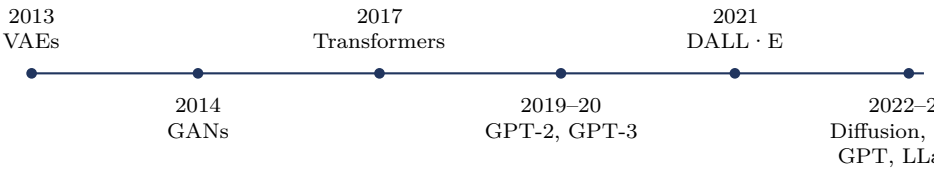
Generative AI feels brand new, but it took roughly a decade of building blocks to reach tools like ChatGPT and DALL · E.

☞ The intuition

Each generation of models added one new trick: a way to compress ideas, a way to compete and sharpen, a way to track long-range patterns, or a way to slowly clean up noise. Modern tools stack all of these tricks together.

★ Everyday picture

Think of cooking. First you learn one fixed recipe card. Then you learn general cooking principles. Later, a trained chef can improvise a whole new dish from a single one-line request. Generative AI followed the same arc, over years of research instead of years of practice.



How to read it: Follow the dots left to right. Each one added a new capability that later models kept and built on – none replaced the one before it entirely.

Year	Milestone	What it added
2013	VAEs	A compressed "idea space" you can nudge to get smooth variations.
2014	GANs	Two networks compete – one fakes, one judges – sharpening realism.
2017	Transformers	Tracked long-range relationships in text far more efficiently.
2019–20	GPT-2, GPT-3	Showed a huge model could write fluent, general-purpose text.
2021	DALL · E	Connected a text prompt directly to a generated image.
2022–23	Diffusion, ChatGPT, LLaMA	Cleaned images out of noise step by step; put chat interfaces and open models into everyday use.

✗ Watch out

It's tempting to think one tool, like ChatGPT, came out of nowhere. It didn't. Each new release reuses ideas from years of earlier research.

✔ Key takeaway

Generative AI is not one invention. It is a stack of ideas – compression, competition, attention, and denoising – built on top of each other over roughly a decade.

3 Under the hood: the stages of a generative AI system

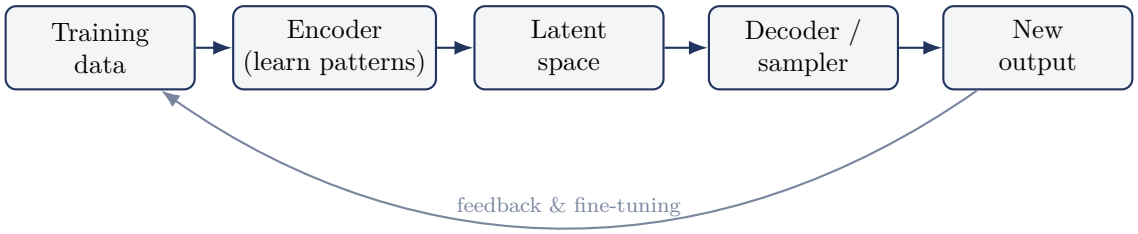
Every generative AI tool, however flashy the output, runs through the same handful of stages behind the scenes.

☞ The intuition

A generative model trains on huge data, squeezes what it learned into a compact "idea map," and then expands a point on that map back out into new content.

★ Everyday picture

Picture a student learning to paint. First, they study thousands of paintings (training). Next, they form a mental sense of what makes a good landscape (the idea map). Then they paint their own new landscape (generation). Finally, they get feedback and adjust (fine-tuning). A generative model follows the same arc.



How to read it: Data flows left to right into a compressed "idea space," then back out as new content. The curved arrow shows the model improving over time from feedback.

The four building-block terms.

- **Encoder** – the part of the model that reads input and squeezes it down into a compact code.
- **Latent space** – the compact code itself: a map of "ideas" where similar things sit close together.
- **Decoder** – the part that takes a code, a point on that idea-map, and expands it back into full content: an image, a sentence, a sound.
- **Sampling** – picking one specific output from the many the model could produce. A little randomness makes each answer slightly different, even for the same prompt.

The four common architecture families.

- **GANs** (Generative Adversarial Networks) – two networks compete. A generator tries to fake realistic content; a discriminator tries to catch the fakes. Both improve together.
- **VAEs** (Variational Autoencoders) – learn a smooth latent space, so nudging the code a little gives a smoothly varied output.
- **Autoregressive models** – generate one piece at a time. Each new piece is chosen based on everything generated so far. This is how GPT-style models write text.
- **Diffusion models** – start from pure random noise and remove a little noise at a time, step by step, until a clean image appears.

✎ Worked example: watching an autoregressive model write one word at a time

Step 1. **Start with the prompt.** The model reads: "Write a two-line poem about rain."

Step 2. **Predict the next word.** It scores every possible next word given everything so far. "The" scores highest.

Step 3. **Sample one word.** It picks "The" (usually the top score, sometimes a close second, to keep answers varied).

Step 4. **Feed that word back in.** The input is now "Write a two-line poem about rain. The" – and step 2 repeats.

Step 5. **Keep going, one word at a time.** "The rain taps softly on the glass," then a line break, then the second line, until the model decides to stop.

Step 6. **Sense-check.** Every full reply you see from a chat model was built one small piece at a time, never all at once.

Where you'll meet this in real tools. GPT-style models are autoregressive. DALL · E 2 and Stable Diffusion are diffusion models. StyleGAN is a GAN. Many face-generation apps use VAEs.

✗ Watch out

Don't confuse "latent space" with literal storage of training images. The model does not keep copies of training photos; it keeps compressed patterns learned from them.

✔ Key takeaway

Every generative model, however different it looks on the outside, does the same job on the inside: compress patterns, then rebuild new content from them.

4 Generative AI at work: applications

The same handful of building blocks power a wide range of everyday tools.

Application	What it does	Example tool
Text generation	Writes coherent, on-topic text from a prompt.	ChatGPT, Claude
Chatbots	Holds a live, natural conversation instead of picking from canned replies.	ChatGPT, virtual assistants
Summarization	Reads long text and writes a shorter version that keeps the key points.	News and research summarizers
Code generation	Turns a plain-English request into working source code.	GitHub Copilot
Image generation	Creates a new image from a text prompt, sketch, or another image.	DALL · E, Midjourney, Stable Diffusion
Speech & audio	Produces human-like speech, cloned voices, or original music.	Voice cloning tools, AI music generators
Video synthesis	Creates or manipulates video, including deepfakes of a real person's face or voice.	AI video-editing tools

✎ Worked example: summarizing a paragraph

Step 1. Start with the full text. "Generative AI models learn patterns from very large datasets. They can then produce new text, images, or audio that resemble what they were trained on. This differs from traditional AI, which typically sorts input into fixed categories rather than creating anything new."

Step 2. Find the one key sentence. Generative AI learns patterns, then creates new content – unlike traditional AI, which only sorts.

Step 3. Drop supporting detail. Remove the examples and the "differs from" clause's extra wording.

Step 4. Write the short version. "Generative AI learns from large datasets and creates new text, images, or audio – unlike traditional AI, which just sorts input."

Step 5. Sense-check. Does the short version still answer "what is this paragraph about?" Yes, in one sentence instead of three.

✎ Worked example: turning a plain request into code

Step 1. Write the request in plain English. "Write a function that adds two numbers."

Step 2. The model reads the intent. It identifies: a function, two inputs, one output, addition.

Step 3. It generates matching code.

```
def add(a, b):
    return a + b
```

Step 4. Sense-check. Call `add(2, 3)`; it should return `5`. The generated code should always be checked before use, the same as code from a junior colleague.

★ Everyday picture

Try it yourself: open Microsoft's free Image) Creator and type a description of anything you can imagine – notice how the wording of your prompt changes the picture you get.

✗ Watch out

Video synthesis and deepfakes are the riskiest application here: they can put real words or actions into a real person's mouth without consent. Section 8 covers this in full.

✔ Key takeaway

The same building blocks – learn patterns, then generate – power everything from a one-line chatbot reply to a fully rendered video.

5 Prompt engineering: talking to AI the right way

A generative model only ever responds to what you actually type. Prompt engineering is the skill of choosing those words well.

☞ The intuition

Prompt engineering is the careful design of the instructions, called a *prompt*, that you give an AI model, so it produces the response you actually want. **Prompting** itself is simply the act of instructing an AI to do a task: you give it a set of instructions, and it carries them out. A prompt can be one short question or a long, detailed brief.

★ Everyday picture

Think of briefing a new intern. Vague instructions get vague work. Specific instructions, with context and an example of what "good" looks like, get exactly what you wanted the first time.

✗ Watch out

The same idea, worded two different ways, can produce very different answers. Word choice matters more with AI than most people expect the first time they try it.

✔ Key takeaway

A prompt is not a search query – it is an instruction. Treat it like a brief you would give a capable but literal-minded assistant.

6 Three techniques that make a prompt better

☞ The intuition

Three habits consistently improve a prompt: refine it over several tries, be explicit about exactly what you want, and give the model a scene or role to stand in.

✎ Worked example: turning a vague prompt into a sharp one (explicitness)

Step 1. **Start with the vague version.** "Tell me about apples."

Step 2. **Ask: what do I actually want to know?** Nutrition? History? Varieties? The vague prompt does not say.

Step 3. **Add the missing detail.** "What are the nutritional benefits of eating apples?"

Step 4. **Compare the two answers.** The vague prompt returns a general essay; the explicit prompt returns a focused list of vitamins, fiber, and calories.

Step 5. **Sense-check.** If you cannot predict roughly what answer you will get, the prompt is not explicit enough yet.

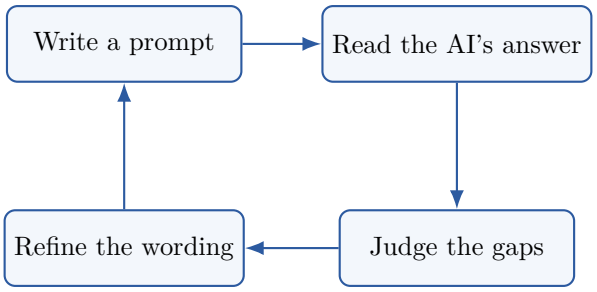
✎ Worked example: giving the AI a scene to stand in (contextual framing)

Step 1. **Start bare.** "Explain the importance of sports."

Step 2. **Add a scene.** "As a teacher, explain to students the importance of sports."

Step 3. **Notice what changes.** The framed version aims the tone and reading level at students, and speaks in a teacher's voice.

Step 4. **Sense-check.** A good frame answers "who is speaking, and to whom?" before the model writes a single word.



How to read it: Iterative refinement is a loop, not a one-shot attempt. Follow the arrows clockwise – each pass should make the prompt a little sharper than the one before.

✗ Watch out

Iterative refinement without explicitness just repeats the same vague question. Refine the *wording*, not just the number of attempts.

✔ Key takeaway

Refine, be explicit, and set the scene – these three habits fix most weak prompts.

7 Five ways to shape a prompt

Style	What you give the model	Best for
Instruction prompting	A direct command, no persona.	Quick, well-defined tasks.
Role prompting	A persona: "You are a ___."	Tasks needing a specific expertise or tone.
Zero-shot	The prompt alone, no examples.	Testing the model's raw ability.
One-shot	The prompt plus one worked example.	Showing an exact format once.
Few-shot	The prompt plus two or more examples.	Teaching a pattern the model should copy.

✎ Worked example: instruction prompting: reformatting a name

Step 1. **Give the raw instruction.** "The name 'Abdul Aziz Md' was entered as first-name-then-last-name, but I don't know which part is which. Give me the name in Lastname Firstname format."

Step 2. **The model identifies the parts.** It reasons that "Md" is likely an abbreviation for "Mohammed," often placed first as a title-like element in some naming conventions.

Step 3. **It rearranges the parts.** Output: "Md Abdul Aziz."

Step 4. **Sense-check.** Always confirm a reformatted name against the original source when it matters, such as on an official document.

✎ Worked example: instruction prompting: redacting personal information

Step 1. **Give the instruction and the text.** "Check the email below for personal identifiable information. If found, replace it with a placeholder – for example, replace 'Abdul Aziz' with [Name]."

Step 2. **The model scans the text.** It looks for names, phone numbers, addresses, and similar details.

Step 3. **It replaces each match.** "Abdul Aziz" becomes "[Name]" everywhere it appears.

Step 4. **It outputs the redacted text.** The rest of the email is left unchanged.

Step 5. **Sense-check.** Re-read the output once yourself – automatic redaction can miss an unusual name format or spelling.

★ Everyday picture

Role prompting example: "You are a doctor. Explain in simple terms why a mild fever is not always dangerous." The same question asked without a role usually comes back more generic and less reassuring in tone.

Shot type	Example prompt shown to the model
Zero-shot	"Classify the sentiment of: 'This movie was absolutely wonderful.'"
One-shot	"Text: 'I loved it.' Sentiment: Positive. Text: 'This movie was absolutely wonderful.' Sentiment:"
Few-shot	"Text: 'I loved it.' Sentiment: Positive. Text: 'It was okay.' Sentiment: Neutral. Text: 'This movie was absolutely wonderful.' Sentiment:"

How to read it: Each row adds more examples than the last. Zero-shot gives no pattern to copy; few-shot gives the model a clear pattern – and a label scale – to follow.

✗ Watch out

Few-shot examples must be balanced and consistent in format, or the model copies the wrong pattern. All-positive examples can quietly bias it toward always answering "Positive."

✔ Key takeaway

Instruction versus role changes *who* is answering; zero-, one-, and few-shot change *how much of a pattern* you hand the model before asking your real question.

8 Guardrails: ethics and pitfalls of generative AI

Creative power comes with responsibility. This section covers the three risks every user of generative AI should recognize.

☞ The intuition

Bias means a model trained mostly on one type of example performs worse, or produces stereotypes, for situations it saw too little of. **Deepfakes** are realistic fake video or audio of a real person doing or saying something they never did. **Copyright issues** arise because some training data includes copyrighted art, text, or music, and who owns the AI's output is still being argued in courts and in law.

★ Everyday picture

Bias is like a photocopier fed mostly one kind of document: it gets very good at that kind, and clumsy at everything else. A deepfake is like a skilled impersonator's voice on the phone – convincing, but not the real person.

✗ Watch out

Never use a generative tool to impersonate a real person without their consent. Always disclose AI-generated content where it matters – news reporting, legal evidence, or academic work.

✔ Key takeaway

Generative AI's power to create is exactly why it needs a human check before publishing – especially anything involving a real person's face, voice, or reputation.

Glossary & key terms

Generative AI	AI that creates brand-new content (text, images, audio, code) rather than only sorting existing content.
Discriminative AI	AI that sorts input into an already-known category, such as ”spam” or ”not spam.”
Prompt	The instruction or question you give an AI model.
Prompt engineering	The careful design of a prompt to get the response you want.
Encoder	The part of a model that compresses input into a compact code.
Decoder	The part of a model that expands a code back into full content.
Latent space	The compact ”idea map” a model builds from its training data.
Sampling	Picking one specific output from the many a model could produce.
GAN	Generative Adversarial Network – two networks compete to sharpen realism.
VAE	Variational Autoencoder – learns a smooth latent space for gradual variation.
Autoregressive model	Generates output one piece at a time, each piece based on everything generated so far.
Diffusion model	Builds an image by removing noise step by step from a random start.
Zero-shot	A prompt with no examples included.
One-shot	A prompt with exactly one worked example included.
Few-shot	A prompt with two or more worked examples included.
Deepfake	A realistic fake video or audio of a real person.
PII	Personally identifiable information – names, addresses, or other details that identify a specific person.

Further reading.

- Goodfellow, I. et al. (2014). *Generative Adversarial Networks.* [arxiv.org](https://arxiv.org/abs/1406.2661)
- Kingma, D.P. & Welling, M. (2013). *Auto-Encoding Variational Bayes.* [arxiv.org](https://arxiv.org/abs/1312.6114)
- Radford, A. et al. (2019). *Language Models are Unsupervised Multitask Learners.* OpenAI.
- Brown, T.B. et al. (2020). *Language Models are Few-Shot Learners (GPT-3).* [arxiv.org](https://arxiv.org/abs/2005.14165)
- Ramesh, A. et al. (2021). *Zero-Shot Text-to-Image Generation (DALL · E).* [arxiv.org](https://arxiv.org/abs/2102.12092)
- Bommasani, R. et al. (2021). *On the Opportunities and Risks of Foundation Models.* [arxiv.org](https://arxiv.org/abs/2108.07258)
- Touvron, H. et al. (2023). *LLaMA: Open and Efficient Foundation Language Models.* [arxiv.org](https://arxiv.org/abs/2302.13971)

- OpenAI (2023). *GPT-4 Technical Report.* [openai.com](https://openai.com/research/gpt-4)